Self-Modification Governance Module (SMGM)

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Canonical Definition

Self-Modification Governance Module (SMGM) defines the structural conditions under which autonomous operational systems may modify their own operational capabilities while preserving governance integrity, operational safety, accountability, and continuous governability.

SMGM governs self-modification.

The module establishes the governance conditions required to ensure that self-modifying autonomous systems remain authorized, controlled, validated, and continuously governed.

Autonomous evolution may include self-modification.

Self-modification requires governance.

SMGM governs that governance.

Module Operational Role

SMGM defines the self-modification governance layer of AEVS.

Its role is to govern autonomous capability changes initiated by the autonomous system itself.

Module Operational Space

SMGM governs:

- self-modification governance,
- autonomous self-adaptation,
- governed self-improvement,
- autonomous capability modification,
- controlled autonomous evolution,
- self-change authorization,
- governed operational adaptation,
- trustworthy self-modification.

Module Function

The module applies wherever autonomous operational systems are capable of modifying their own operational behavior, decision logic, parameters, or capabilities.

Minimum Implementation Framework

1. Define the Self-Modification Object

Identify which autonomous capabilities may perform self-modification.

2. Define Self-Modification Conditions

Establish governance conditions governing when and how self-modification is permitted.

3. Define Self-Modification Failure Logic Identify conditions where self-modification becomes unauthorized, unsafe, uncontrolled, unverifiable, or governance- invalid.

4. Define Governance Response

Define governance actions for approval, restriction, rollback, escalation, suspension, or additional validation.

5. Preserve Self-Modification Traceability

Preserve self-modification requests, governance decisions, operational evidence, modification history, and supporting documentation.

Use Case 1 — Autonomous Industrial Robot

Scenario

An autonomous production robot identifies an opportunity to improve its task execution efficiency.

Application

SMGM governs whether the proposed self-modification satisfies predefined governance conditions before implementation.

Result

Operational improvement remains controlled, validated, and governance-approved.

Use Case 2 — Enterprise Autonomous AI Agent

Scenario

An autonomous AI agent proposes modifications to its workflow optimization strategy based on operational experience.

Application

SMGM evaluates the proposed self-modification according to governance rules before activation.

Result

The organization preserves safe, accountable, and continuously governed autonomous evolution.

Canonical Closing Statement

**Governable evolution requires governable self-modification.
Self-Modification Governance ensures that autonomous operational systems improve themselves only through authorized, controlled, traceable, and continuously governed evolution.**

Related Documents

→ Autonomous Evolution Standard (AEVS)

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Canonical Source

https://originopenfoundation.org/